

Generation Interconnection System Impact Study Report

For Queue Project Q20-1417

Wells River 161 kV

July 23, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1417. The Interconnection Customer proposes a 95.8 MW solar photovoltaic facility interconnecting at the Wells River 161 kV substation in the New York area. The request was received on March 04, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 464 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,386,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Belfort 138 kV/1 kV Auto

Contingency: Salton Sea 2 / Gastonia Prime Power Park 138.0 kV Line

Loading: 116.3% (Rating: 100 MVA)

DFAX: 6.5%

Pre-project loading: 110.1%

2. Facility: Gastonia Prime Powe Park / Wells Riv 161.0 kV

Contingency: Closter / Lee Vining 161.0 kV

Loading: 102.8% (Rating: 361 MVA)

DFAX: 57.1%

Pre-project loading: 87.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Belfort 138 kV/1 kV Auto	Replace transformer with higher-rated unit	\$5,196,000	100% (\$5,196,000)
Gastonia Prime Powe Park / Wells Riv 161.0 kV	Reconductor line	\$324,000	59% (\$190,000)
	Total allocated to project		\$5,386,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.